

# JEE MAIN REVISION MASTER: SURFACE CHEMISTRY OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 113 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                         | Frequency | Category        |
|---------------|--------------------------------------|-----------|-----------------|
| Very High     | Surface Chemistry - Core Application | 113 PYQs  | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Surface Chemistry - Core Application (Very High Yield — 113 PYQs)

**Core Principle:** Key problem pattern in Surface Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

**Representative Source PYQs:**

- [2024 Evening Shift] Q#1 (MCQ): "The adsorbent used in adsorption chromatography is/are - A. silica gel B. alumina C. quick lime D. magnesia Choose the most appropriate answ..." [Ans: N/A]
- [2024 Morning Shift] Q#2 (MCQ): "'Adsorption' principle is used for which of the following purification method?..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Four gases A, B, C and D have critical temperatures 5.3,33.2,126.0 and 154.3K respectively. For their adsorption on a fixed amount of ch..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: JEE Main 2023 (Online) 12th April Morning Shift (Q#1)**

Four gases A, B, C and D have critical temperatures 5.3, 33.2, 126.0 and 154.3 K respectively. For their adsorption on a fixed amount of charcoal, the correct order is:

- (A) > C > B > A
- (B) > B > D > A
- (C) > C > A > B
- (D) > D > B > A

**Correct Answer: None**

**Solution Analysis:** The extent of adsorption of a gas on a solid surface depends on the critical temperature of the gas and the temperature of the solid surface. The critical temperature is the temperature above which a gas cannot be liquefied, no matter how much pressure is applied. Therefore, gases with higher critical temperatures are less likely to be adsorbed on a solid surface than gases with lower critical temperatures. In this case, the gases are A, B, C, and D, with critical temperatures of 5.3 K, 33.2 K, 126.0 K, and 154.3 K, respectively. Therefore, the correct order of adsorption on a fixed amount of